



Metasurfaces on silicon photonic waveguides for simultaneous emission phase and amplitude control

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Abstract: Chip-scale photonic systems that manipulate free-space emission have recently attracted attention for applications such as free-space optical communications and solid-state LiDAR. Silicon photonics, as a leading platform for chip-scale integration, needs to offer more versatile control of free-space emission. Here we integrate metasurfaces on silicon photonic waveguides to generate free-space emission with controlled phase and amplitude profiles. We demonstrate experimentally structured beams, including a focused Gaussian beam and a Hermite-Gaussian TEM₁₀ beam, as well as holographic image projections. Our approach is monolithic and CMOS-compatible. The simultaneous phase and amplitude control enable more faithful generation of structured beams and speckle-reduced projection of holographic images.

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Silicon photonics enables monolithic integration of various functional components to form chip-scale photonic systems. It has recently shown promise in applications that involve the manipulation of free-space emission [1], including free-space optical communications (FSO) [2–5], light detecting and ranging (LiDAR) [6–8], laser-scanning microscopy [9], trapped-ion quantum computation [10,11], optogenetics [12], and display [13]. These applications often require versatile control of the free-space emission. For example, highly collimated beams are needed for LiDAR to achieve long detection ranges [2,6,14]. Focused beams are used in microscopy and quantum computation to optically address a small volume in space [9,11]. Orbit-angular-momentum beams and other structured waves are used to boost the communication channels for FSO [5]. Various approaches have been developed on the silicon photonic platform to generate free-space emission with tailored phase profiles. One type of approach uses apodized shallow-etched surface gratings [15,16]. Another type of approach is based on monolithic integration of metasurfaces on silicon waveguides [17–25]. Because each subwavelength-spaced meta-atom can control the local phase value, metasurfaces allow more abrupt manipulation of the phase profiles than surface gratings. Metasurfaces integrated on waveguides also enable more functionalities, such as polarization control [23,24] and waveguide mode conversion [25]. Experimental demonstrations of integrated metasurfaces on silicon photonic waveguides for emission control have been reported in recent literature, including Ref. [18,19], which use resonant Au-dielectric-Au sandwiched nanobars as meta-atoms, and our previous work Ref. [17], in which non-resonant silicon nanopillars are employed. In our previous work, we leveraged the robust phase control of the non-resonant metasurfaces to demonstrate experimentally diffraction-limited beam focusing with a Strehl ratio of 0.82, as well as a complicated phase profile that generates holographic projection [17]. The demonstrated metasurfaces in that work are monolithically integrated and compatible with the complementary metal-oxide-semiconductor (CMOS) processes.

Despite the successful tailoring of the emission phase profiles demonstrated in the previous metasurface works [17–25], there is a need for simultaneous control of the phase and amplitude profiles. Because a field is generally complex-valued, to faithfully form a structured beam, such as a Hermite-Gaussian beam or a Bessel beam, requires reproducing both the phase and amplitude profiles [26,27]. Simultaneous phase and amplitude control are also imperative for holography because a phase-only hologram can only form an approximated image with speckle noises [28,29]. To control the phase and amplitude independently, two degrees of freedom are needed. In this paper, we vary the positions and diameters of silicon nanopillars, in contrast to our previous work, where the positions are fixed to a square lattice [17]. We demonstrate the generation of a focused Gaussian beam, a Hermite-Gaussian beam, and holographic projections with reduced speckles. The demonstrated metasurfaces are monolithically integrated on silicon waveguides, and the materials are all CMOS-compatible.

We monolithically integrate metasurfaces on silicon photonic waveguides and leverage the meta-atom position and size to control the emission phase and amplitude, respectively. The metasurfaces are composed of amorphous silicon (a-Si) nanopillars with heights of 500 nm. These meta-atoms are integrated on standard 220 nm-thick silicon photonic waveguides, as shown in Fig. 1. The operating wavelength λ is 1550 nm. We design the metasurfaces to couple perturbatively to the guided TE_0 mode of the waveguide. To ensure the perturbative condition, we design a 50 nm SiO_2 layer between the silicon waveguide and the metasurface, which also serves as the etching hard mask.

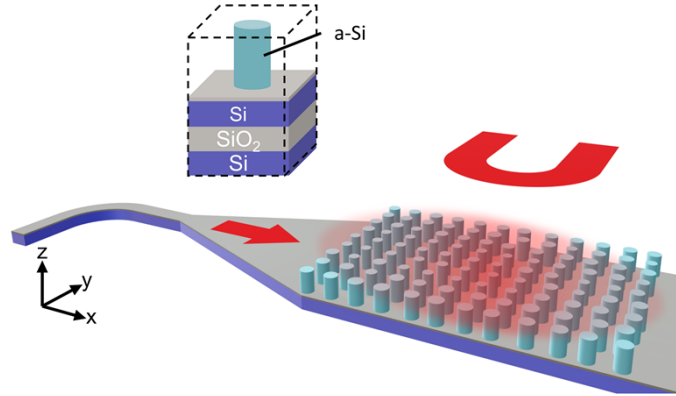


Fig. 1. Schematic of the integrated metasurface. The metasurface is composed of 500 nm-tall a-Si nanopillars, which are monolithically integrated on top of a 220 nm-thick silicon photonic waveguide. There is a 50 nm SiO_2 layer between the silicon waveguide and the metasurface. Below the 220 nm-thick silicon layer are the 3 μm -thick buried oxide (SiO_2) and the silicon handle wafer.

We leverage the meta-atom positions to control the emission phase profile. The a-Si nanopillars are equally spaced by 600 nm in the y direction. In contrast, we vary the spacing in the x direction, i.e., the wave propagation direction, to control the emission phase. As illustrated in Fig. 2(a), because a meta-atom perturbs the waveguide mode and out-couple the light at its position, the emitting field inherits the phase of the underlying waveguide mode at this particular position. If we place a meta-atom at position (x_i, y_i) , the phase of the emitting field at this position, denoted by $\phi_t(x_i, y_i)$, is determined by

$$\phi_t(x_i, y_i) = \phi_{\text{WG}}(x_i, y_i) + \phi_{\text{MS}}(d_i(x_i, y_i)), \quad (1)$$

where $\phi_{\text{WG}}(x, y) = \beta x = 2\pi x / (\lambda / n_{\text{eff}})$ is the propagation phase of the waveguide mode. β is the propagation constant, and n_{eff} is the effective refractive index of the TE_0 mode. An

accurate n_{eff} value that takes into account the presence of the nanopillars can be obtained by the finite-difference time-domain (FDTD, Lumerical Inc.) simulation [17]. We plot $\phi_{\text{WG}}(x, y)$ in Fig. 2(a) and wrap it every 2π . It changes linearly from 0 to 2π repeatedly with a period of λ/n_{eff} . $\phi_{\text{MS}}(d_i)$ is the phase shift introduced by the meta-atom, which is determined by the diameter d_i of the nanopillar at position (x_i, y_i) . In the first design step, we fix the nanopillar diameter. Therefore, we can neglect the ϕ_{MS} term in Eq. (1) because it is only an irrelevant constant. To generate an emission with a target phase profile $\phi_t(x, y)$, we can find the intersection between $\phi_t(x, y)$ and $\phi_{\text{WG}}(x, y)$, as illustrated in Fig. 2(a). Then we place meta-atoms at the intersection positions, as denoted by (x_i, y_i) . This procedure allows sampling the target emission phase profile $\phi_t(x, y)$ with subwavelength spacings, creating an emission with the desired target phase profile.

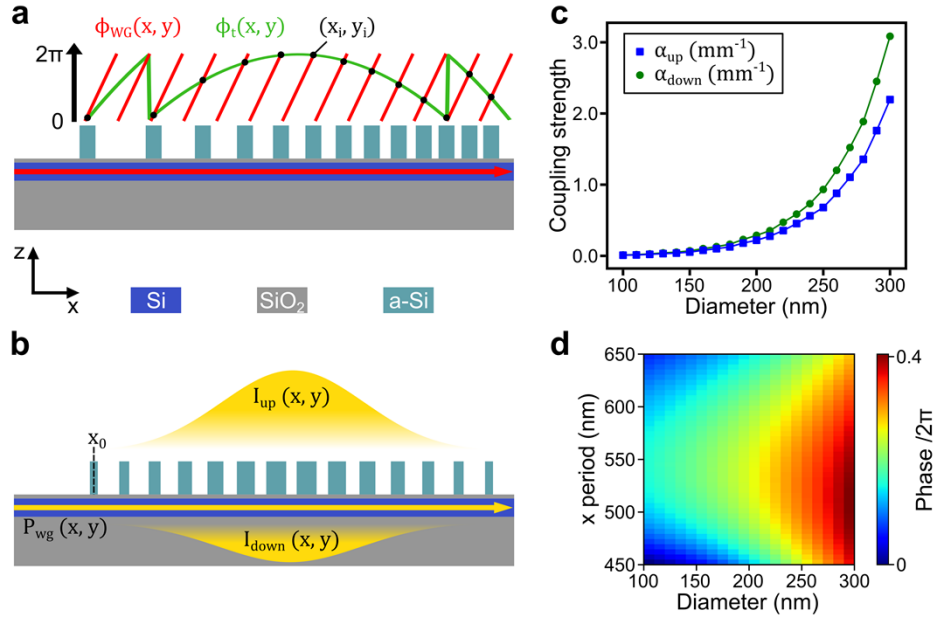


Fig. 2. (a) Schematic of the phase control mechanism by leveraging the positions of the meta-atoms. (b) Schematic of the energy flow in the device, where we leverage the meta-atom sizes to control the intensity profile $I_{\text{up}}(x, y)$. (c) Coupling strength as a function of the meta-atom diameter. (d) Simulated phase shift introduced by the meta-atom as a function of the x period and the diameter of the meta-atom.

We vary the diameters of meta-atoms to control the emission amplitude profile. Figure 2(b) illustrates the energy flow in the device. As the light propagates along the x direction, it starts to encounter the perturbations from the meta-atoms at $x = x_0$. The power in the waveguide starts decaying because of the upward and downward out-coupling. $P_{\text{wg}}(x, y)$ denotes the power per unit y length in the waveguide. According to energy conservation, the relation between upward emission intensity profile $I_{\text{up}}(x, y)$, downward emission intensity profile $I_{\text{down}}(x, y)$, and $P_{\text{wg}}(x, y)$ is given by

$$-\frac{\partial P_{\text{wg}}(x, y)}{\partial x} = I_{\text{up}}(x, y) + I_{\text{down}}(x, y). \quad (2)$$

We define $\alpha_{\text{up}}(x, y)$ and $\alpha_{\text{down}}(x, y)$ as the upward and downward coupling strength, respectively [30]. The relation between the intensity profiles and coupling strengths is given by

$$I_{\text{up}}(x, y) = 2\alpha_{\text{up}}(x, y)P_{\text{wg}}(x, y), \quad (3)$$

$$I_{\text{down}}(x, y) = 2\alpha_{\text{down}}(x, y)P_{\text{wg}}(x, y). \quad (4)$$

We can vary the meta-atom diameter to control the coupling strengths. We perform 3D FDTD simulation to obtain α_{up} and α_{down} as a function of the meta-atom diameter. The meta-atoms are arranged in a rectangular lattice with periods of 546 nm and 600 nm in the x and y directions, respectively. As shown in Fig. 2(c), large meta-atoms create stronger perturbation and, thus, larger coupling strengths. By changing the diameter from 100 nm to 300 nm, we can tune α_{up} from 0.01 mm^{-1} to 2.20 mm^{-1} . By assigning the distribution of meta-atom diameters in the metasurface layout, we can control the $\alpha_{\text{up}}(x, y)$ at different positions and thus apodize the upward emission intensity profile. The large α_{up} tuning range by a factor of 220 indicates a large design window to create various desired intensity profiles.

Here we are interested in creating a target upward emission intensity profile $I_{\text{up}}(x, y)$. Given a target $I_{\text{up}}(x, y)$, we can solve for the needed coupling strength $\alpha_{\text{up}}(x, y)$ from Eqs. (2)–(4).

$$\alpha_{\text{up}}(x, y) = \frac{I_{\text{up}}(x, y)/2}{P_{\text{wg}}(x_0, y) - \int_{x_0}^x [(1 + \alpha_{\text{down}}(x', y)/\alpha_{\text{up}}(x', y)) \times I_{\text{up}}(x', y)] dx'}. \quad (5)$$

Although $\alpha_{\text{up}}(x, y)$ appears on both sides of Eq. (5), this equation can be solved iteratively because the factor $\alpha_{\text{down}}(x, y)/\alpha_{\text{up}}(x, y)$ is known for any given meta-atom diameters (see Fig. 2(c)). After obtaining the required $\alpha_{\text{up}}(x, y)$ with Eq. (5), we use Fig. 2(c) as a look-up table to decide the diameters $d_i(x_i, y_i)$ of all meta-atoms, which determine the initial metasurface layout. We can also calculate the fraction of power coupled upward to the air side by

$$\eta_{\text{up}} = \int I_{\text{up}}(x, y) dx dy / \int P_{\text{wg}}(x_0, y) dy. \quad (6)$$

As we vary the meta-atom diameters to control the intensity profile, the meta-atoms with different diameters also introduce different phase shift. Therefore, we need to further correct the phase shift by slightly repositioning the meta-atoms. Here we perform rigorous coupled-wave analysis (RCWA) to calculate the phase shift as a function of the meta-atom diameter and the period in the x direction (see Fig. 2(d)) [31], which is later used for phase correction.

We summarize the design procedure with the flowchart shown in Fig. 3. (i) We use the target phase profile $\phi_t(x, y)$ to determine the initial meta-atom positions (x_i, y_i) , as depicted in Fig. 2(a). The meta-atom diameter is fixed at 200 nm at this point. (ii) We use the target intensity profile $I_{\text{up}}(x, y)$ to find the required coupling strength $\alpha_{\text{up}}(x, y)$ by applying Eq. (5). The obtained $\alpha_{\text{up}}(x, y)$ is converted to the distribution of meta-atom diameters $d_i(x_i, y_i)$ using Fig. 2(c). (iii) We reposition the meta-atoms slightly using Fig. 2(d) to correct the phase shift introduced by different meta-atom diameters. This step generates the final metasurface layout. Notice that the design procedure relies on the fact that the metasurfaces couple to the underlying guided mode only perturbatively, and the mode purity is well-maintained, which is confirmed with 3D FDTD simulation (see Supplement 1).

We fabricate the devices using a CMOS-compatible method, as shown in Fig. 4(a). We first spin-coat electron-beam photoresist ZEP-520A on a silicon-on-insulator (SOI) wafer. The wafer has a 220 nm top silicon layer and a 3 μm buried oxide layer. Then we use electron beam lithography (EBL) to define the waveguide patterns. We deposit 50 nm of SiO_2 using electron-beam evaporation, followed by the lift-off process to convert the pattern into the hard mask. Next, plasma-enhanced chemical vapor deposition (PECVD) is employed to deposit 500 nm a-Si and 150 nm SiO_2 . We spin-coat electron-beam photoresist ma-N 2403 and define the metasurface pattern using EBL. After that, we use inductively-coupled plasma reactive-ion etching (ICP-RIE) with $\text{C}_4\text{F}_6/\text{Ar}/\text{O}_2$ -based gases to transfer the metasurface pattern to the SiO_2 hard mask. The thickness of the SiO_2 hard mask is chosen according to the etching selectivity and monitored by ellipsometer. Finally, a single etch step with HBr/Cl_2 -based gases is used to finish the metasurface and waveguide structure. The scanning electron microscope (SEM) images of the fabricated devices are shown in Fig. 4(b) and 4(c), respectively.

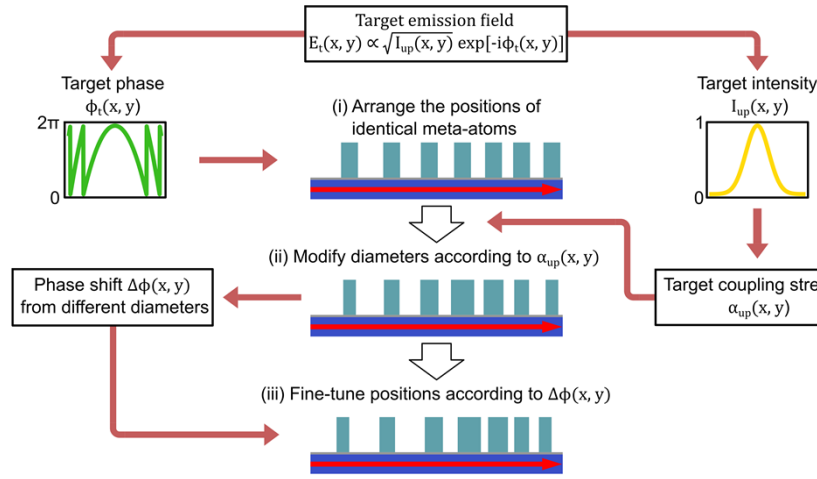


Fig. 3. Flowchart to design a metasurface with simultaneous phase and amplitude control.

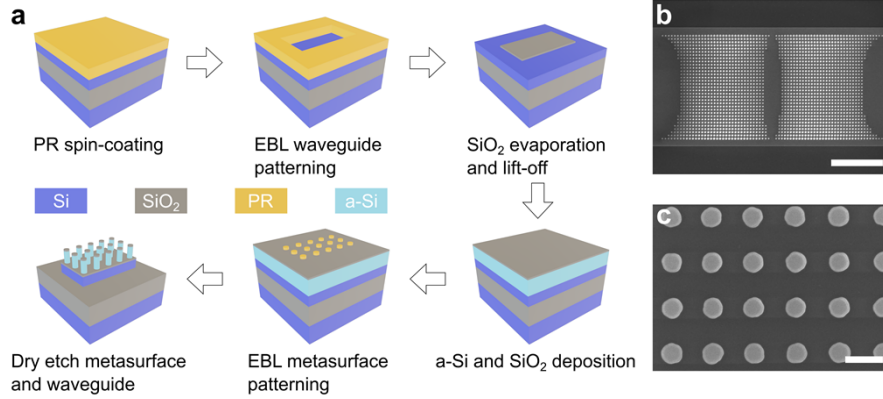


Fig. 4. (a) Schematic of the fabrication processes. (b) SEM image of the fabricated metasurface on a silicon waveguide. (c) Zoom-in SEM image of a-Si nanopillars. The scale bars in (b) and (c) are 10 μm and 500 nm, respectively. PR: photoresist.

We experimentally demonstrate different structured beams and holographic projections to show the capability of simultaneous phase and amplitude control. The first demonstration is the emission of a beam with a Gaussian intensity profile I_G and a hyperbolic focusing phase profile ϕ_G , as given by

$$I_G(x, y, 0) = \exp\left(-2\frac{x^2 + y^2}{w^2}\right), \quad (7)$$

$$\phi_G(x, y, 0) = -k_0 \left(\sqrt{x^2 + y^2 + F^2} - F \right), \quad (8)$$

where w and F are designed to be 7.5 μm and 50 μm , respectively. The plane $z = 0$ is the emission plane right above the metasurface. We design the metasurface following the procedure shown in Fig. 3. The fraction of upward power η_{up} , as defined by Eq. (6), is designed to be 2.5%. We verify the design by performing the FDTD simulation, as shown in Fig. 5(b–d). Figure 5(b) shows the intensity profile on the xz plane. The beam focuses at $z = 35 \mu\text{m}$, which deviates from $F = 50 \mu\text{m}$ because of the finite w value. This can be understood from the paraxial ABCD matrix calculation,

which predicts the approximated focused position to be $F/[1 + (F/z_R)^2]$, where $z_R = \pi w^2/\lambda$ is the Rayleigh range before focusing [32]. Figure 5(c) shows the simulated intensity profile at the emission plane $z=0$. We also compare the simulated intensity profile at the emission plane with the target intensity profile $I_G(x, y, 0)$, showing good agreement (Fig. 5(d)). Notice that the simulated intensity profile in Fig. 5(d) shows weak oscillations near the peak. Because of the discrete nature of the subwavelength-spaced emitters, the near-field intensity distribution is not entirely continuous and shows oscillations with high spatial frequencies. However, these high-frequency oscillations cannot propagate far away from the metasurface and gradually disappear. Next, we follow the process shown in Fig. 4 to fabricate the integrated metasurface on a 22 μm -wide silicon waveguide. A slow taper with a length of 800 μm is used to expand the waveguide width from 450 nm to 22 μm to ensure that no higher-order modes are excited. We measure the fabricated device with the setup shown in Fig. 5(a). The setup uses a tunable laser (Toptica CTL1550) with the wavelength set to 1550 nm. The incident light is coupled into the chip through a grating coupler. To measure the emission, we build an infinite-conjugate imaging system with a 50X NA 0.65 objective (Mitutoyo M Plan Apo NIR) and a near-infrared (NIR) camera (Xenics Bobcat 320). The imaging system is translated by a motorized stage to record intensity profiles at different heights. The measurement results are shown in Fig. 5(e–g). We observe a focused spot at $z = 37 \mu\text{m}$ in Fig. 5(e), which is close to the simulation. Figure 5(f) and (g) show that we successfully generate a Gaussian intensity profile at the emission plane, in good agreement with the target profile $I_G(x, y, 0)$.

Next, we demonstrate the emission of a Hermite-Gaussian TEM_{10} beam. The target emission intensity profile I_{HG} and phase profile ϕ_{HG} are given analytically by

$$I_{\text{HG}}(x, y, z) = \left[\frac{w_0}{w(z - z_0)} \right]^2 \left[\frac{2\sqrt{2}x}{w(z - z_0)} \right]^2 \exp \left[\frac{-2(x^2 + y^2)}{w^2(z - z_0)} \right], \quad (9)$$

$$\phi_{\text{HG}}(x, y, z) = k_0(z - z_0) - 2 \tan^{-1} \left(\frac{z - z_0}{z_R} \right) + \frac{k_0(x^2 + y^2)}{2R(z - z_0)}, \quad (10)$$

where w_0 is the beam waist, $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$ is the z -dependent beam size, $z_R = \pi w_0^2/\lambda$ is the Rayleigh range, and $R(z) = z[1 + (z_R/z)^2]$ is the z -dependent radius of curvature. We design w_0 to be 10 μm and the beam waist to locate at 100 μm above the metasurface, i.e., $z_0 = 100 \mu\text{m}$. The fraction of upward power η_{up} is designed to be 1%. We design the corresponding integrated metasurface and perform the FDTD simulation to verify the design. Figures 6(a) shows the simulated intensity profile on the xz plane, demonstrating the emission of a Hermite-Gaussian TEM_{10} beam from the metasurface. In the simulation, we observe minor deviations from the target intensity profile $I_{\text{HG}}(x, y, z)$: the peak intensity occurs at $z = 68 \mu\text{m}$ instead of the waist position at $z = 100 \mu\text{m}$, and there exist fine structures in the beam near the metasurface. The deviation is due to the fact that the beam emits from a $60 \mu\text{m} \times 20 \mu\text{m}$ metasurface. Therefore, minor diffraction is introduced by truncating the Hermite-Gaussian beam with a $60 \mu\text{m} \times 20 \mu\text{m}$ aperture (see Supplement 1 materials). In Fig. 6(b), we show the simulated xy intensity profile on the emission plane $z = 0$. The simulated intensity profile matches well with the target I_{HG} , as demonstrated by the cross-sectional profile in Fig. 6(c). We fabricate the designed device and perform the measurement. Figures 6(d) and 6(e) show the measured intensity profiles on the xz plane and on the emission plane $z = 0$, respectively. Figure 6(f) compares the measured cross-sectional profile with the target I_{HG} at the emission plane. Our measurements demonstrate the successful generation of the Hermite-Gaussian TEM_{10} beam from the integrated metasurface, while the deviations from the simulation are attributed to the fabrication imperfections.

As a third demonstration, we project holographic images with the integrated metasurfaces. With a $40 \mu\text{m} \times 40 \mu\text{m}$ metasurface on top of a 42 μm -wide Si waveguide, we design a target image

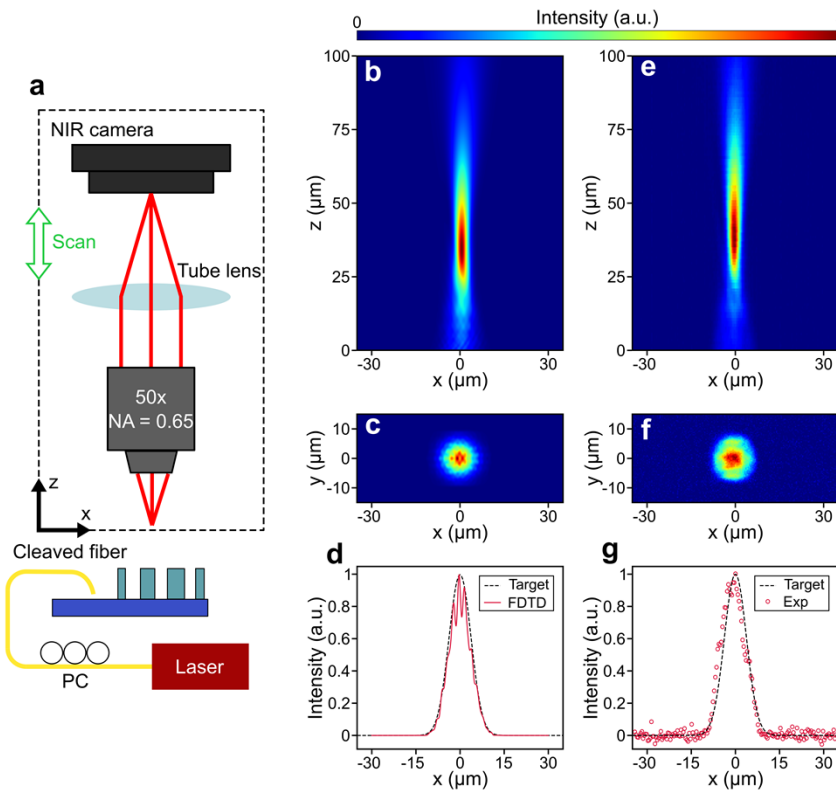


Fig. 5. Simulation and measurement of the focused Gaussian beam emitted from the integrated metasurface. (a) Schematic of the measurement setup. (b) Simulated xz intensity profile. (c) Simulated intensity profile at the emission plane $z = 0$. (d) Comparison between the simulated intensity profile at the emission plane with the target Gaussian distribution. (e) Measured xz intensity profile. (f) Measured intensity profile at the emission plane. (g) Comparison between the measured intensity profile at the emission plane with the target Gaussian distribution. Panel (d) and (g) are cross-sections along $(x, 0, 0)$. PC: polarization controller. Exp: experiment.

of the letter “U” at $30\text{ }\mu\text{m}$ above the metasurface. To show the advantages of simultaneous control of phase and amplitude, we design two types of meta-holograms for comparison: the phase-only (PO) meta-hologram and the phase-amplitude (PA) meta-hologram. The PO meta-hologram controls only the phase profile using the nanopillar positions. All the nanopillars maintain a fixed diameter of 200 nm . We use the Gerchberg-Saxton algorithm to obtain the emission phase profile needed for producing the target image [33], as shown in Fig. 7(a). The SEM image of the fabricated PO meta-hologram is shown in Fig. 7(b). We show the FDTD-simulated and measured results of the PO meta-hologram in Fig. 7(c) and (d). The simulated and measured images locate at $z = 30\text{ }\mu\text{m}$ and $35\text{ }\mu\text{m}$, respectively. Although the measurement agrees well with the simulation, the target image “U” shows many speckles. This is because the Gerchberg-Saxton algorithm can only approximate an image due to the lack of amplitude control. On the other hand, we design the PA meta-hologram that leverages the simultaneous control of phase and amplitude. Instead of the Gerchberg-Saxton algorithm, here we use the band-limited angular spectrum method to back-propagate the binary mask “U” from the image plane to the emission plane [34], which gives the needed emission phase and intensity profiles (see Fig. 7(e-f)). The SEM image of the fabricated PA meta-hologram is shown in Fig. 7(g), where one can see the nanopillars

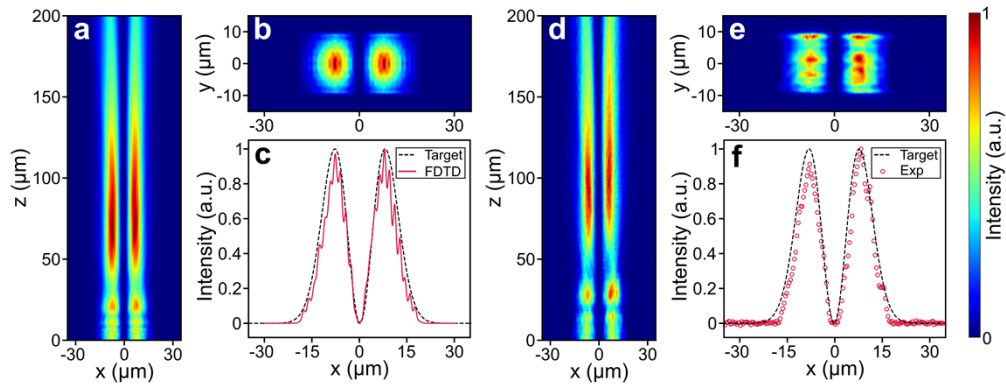


Fig. 6. Simulated and measured intensity profiles of the Hermite-Gaussian TEM_{10} beam emitted from the integrated metasurface. (a) Simulated xz intensity profile. (b) Simulated intensity profile at the emission plane $z = 0$. (c) Comparison between the simulated intensity profile at the emission plane with the target. (d) Measured xz intensity profile. (e) Measured intensity profile at the emission plane. (f) Comparison between the measured intensity profile at the emission plane with the target. Panel (c) and (f) are cross-sections along $(x, 0, 0)$. Exp: experiment.

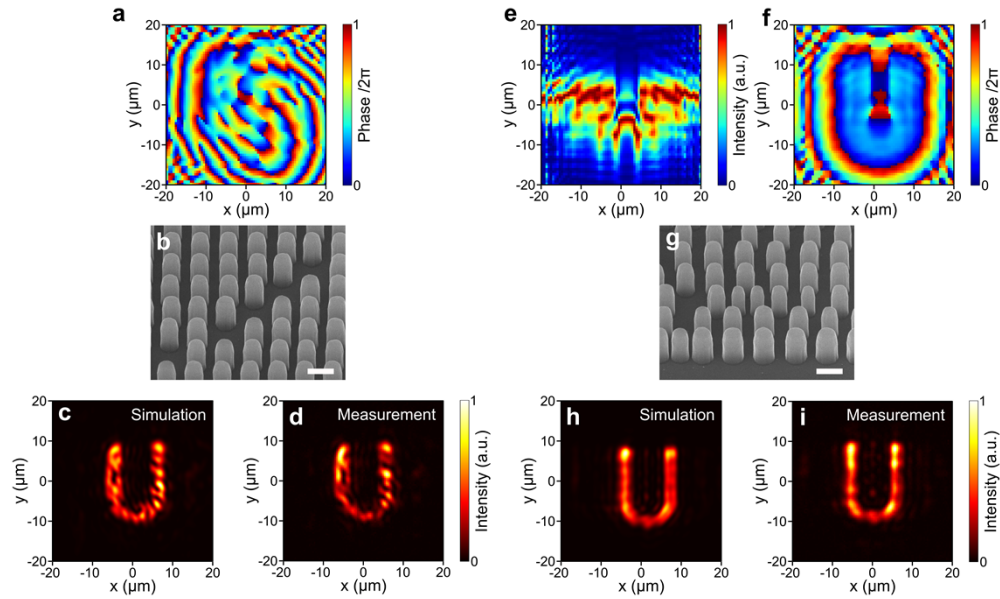


Fig. 7. Holographic projection from the integrated metasurfaces. (a) Target phase profile obtained using the Gerchberg-Saxton algorithm for the phase-only (PO) meta-hologram. (b) Tilted SEM image of the fabricated PO meta-hologram. (c-d) Simulated and measured holographic images of the letter 'U' projected from the PO meta-hologram. (e-f) Target intensity and phase profiles used in the phase-amplitude (PA) meta-hologram. (g) Tilted SEM image of the fabricated PA meta-hologram. (h-i) Simulated and measured holographic images of the letter 'U' projected from the PA meta-hologram. The scale bars in (b) and (g) are 500 nm.

have different diameters. Figure 7(h) and (i) are the simulated and measured holographic images, located at $z = 30\ \mu\text{m}$ and $37\ \mu\text{m}$, respectively. The holographic image generated with PA meta-hologram has sharper boundaries and fewer speckle noises than that generated with the PO meta-hologram.

In conclusion, we have developed metasurfaces that can be integrated monolithically on silicon photonic waveguides to create free-space emission with controlled phase and amplitude profiles. The phase and amplitude are controlled via the position and size of the meta-atoms, respectively. We demonstrate experimentally the generation of structured beams, including the focused Gaussian beam and the Hermite-Gaussian TEM_{10} beam, as well as the projection of holographic images with reduced speckle noises. In current demonstrations, the fraction of upward power η_{up} is limited by the metasurface length in the propagation direction, i.e., x direction. The maximum coupling strength α_{up} in the current meta-atom library is $2.20\ \text{mm}^{-1}$, indicating that the metasurface needs a length of millimeter scale to out-couple most of the optical power. Therefore, the current library is more suitable for making larger-area metasurfaces if a higher η_{up} is needed. One can increase the coupling strength by choosing larger nanopillar diameters or by reducing the thickness of the SiO_2 layer between the a-Si nanopillar and the Si waveguide. The metasurfaces we developed offer CMOS compatibility and monolithic integration with silicon photonics. The integrated metasurfaces can be applied to chip-scale photonic systems that require precise manipulation of free-space emission.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See [Supplement 1](#) for supporting content.

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